

NERC

NORTH AMERICAN ELECTRIC
RELIABILITY CORPORATION

2025 Frequency Response Annual Analysis

August 2025

This report was approved by the Resources Subcommittee on August 13, 2025.

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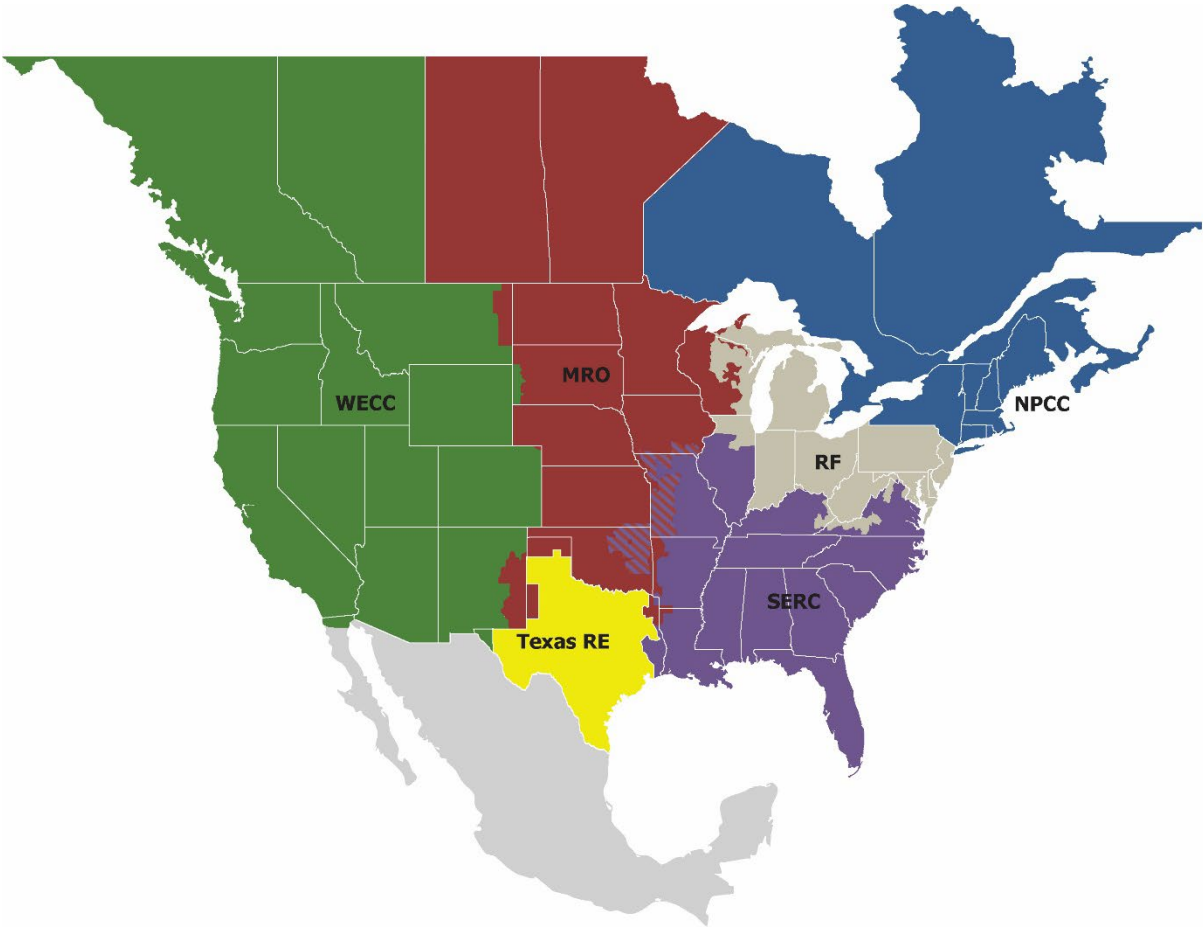
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Preface

Electricity is a key component of the fabric of modern society and the Electric Reliability Organization (ERO) Enterprise serves to strengthen that fabric. The vision for the ERO Enterprise, which is comprised of NERC and the six Regional Entities, is a highly reliable, resilient, and secure North American bulk power system (BPS). Our mission is to assure the effective and efficient reduction of risks to the reliability and security of the grid.

Reliability | Resilience | Security
Because nearly 400 million citizens in North America are counting on us

The North American BPS is made up of six Regional Entities as shown on the map and in the corresponding table below. The multicolored area denotes overlap as some load-serving entities participate in one Regional Entity while associated Transmission Owners/Operators participate in another.



MRO	Midwest Reliability Organization
NPCC	Northeast Power Coordinating Council
RF	ReliabilityFirst
SERC	SERC Reliability Corporation
Texas RE	Texas Reliability Entity
WECC	WECC

Executive Summary

This report is the 2025 annual analysis of frequency response performance for the administration and support of *NERC Reliability Standard BAL-003-2 – Frequency Response and Frequency Bias Setting*,¹ effective December 1, 2020. It updates the statistical analyses and calculations contained in the *2012 Frequency Response Initiative Report*² that was approved by the NERC Resources Subcommittee (RS) and the technical committee, which predated the Reliability and Security Technical Committee (RSTC), and was accepted by the NERC Board of Trustees (Board).

This report is prepared by NERC staff³ and contains the annual analysis, calculation, and recommendations for the interconnection frequency response obligation (IFRO) for each of the four electrical Interconnections of North America for the operating year (OY) 2026 (December 2025 through November 2026). Below are the key findings and recommendations contained in this report.

Key Findings

Starting Frequency

The starting frequency for the calculation of IFROs, shown in [Table 1.1](#), is the fifth percentile of the five-year probability distribution of the respective interconnection frequency, representing a 95% chance that frequencies will be at or above that value at the start of any frequency event. The starting frequency remained the same for the Eastern Interconnection (EI) at 59.971 Hz, the Western Interconnection (WI) at 59.970 Hz, and the Texas Interconnection (TI) at 59.970 Hz. However, the Québec Interconnection (QI) starting frequency decreased slightly from 59.965 Hz to 59.964 Hz.

Frequency Probability Density Functions

The standard deviation is a measure of the dispersal of frequency values around the mean value; a smaller standard deviation indicates tighter concentration around the mean value and more stable performance of Interconnection frequency. Analysis of the frequency probability density functions shows that standard deviations have been flat (Eastern and Western) or fluctuating within a small range (Texas and Québec). Comparisons of annual frequency profiles for each Interconnection are shown in [Figure 1.6](#), [Figure 1.7](#), [Figure 1.8](#), and [Figure 1.9](#).

Interconnection Performance and the Comparison of Mean Value A, B, and Point C

[Table 2.6](#) shows a comparison of mean Value A, mean Value B, and mean Point C that is illustrative of Interconnection performance during low-frequency events over the previous OY and as compared to the 2016 OY in which the IFRO values were frozen. Loss-of-load events have been excluded from the data in [Table 2.6](#). The Eastern, Western, and Texas Interconnections show an increase in mean Value B and a decrease in the mean (A-B), indicating improved performance during the stabilizing period of frequency events. Québec showed an increase in mean Value B up until OY 2024, where it declined slightly. The Western, Texas, and Québec Interconnections show an increase in mean Point C as well as a decrease or no change in mean (A-C), indicating improved performance during the arresting period of frequency events. The Eastern Interconnection shows no change in mean Value C and a slight decrease in mean Value A-C. This performance data demonstrates that the higher calculated IFROs are due to improved stabilizing period performance and not due to a decline in the performance of the Point C nadir.

¹ [Standard BAL-003-2](#)

² "Frequency Response Initiative Report: The Reliability Role of Frequency Response," 2012. Available: https://www.nerc.com/pa/Stand/Project%20200712%20Frequency%20Response%20DL/FRI_Report_10-30-12_Master_w-appendices.pdf

³ Prepared by the NERC Advanced System Analytics and Modeling department.

Recommendations

NERC provides the following recommendation for the administration of *Standard BAL-003-2*¹ for OY 2026 (December 1, 2025, through November 30, 2026):

- The IFRO value for the TI will change by **-4 MW/0.1 Hz** due to a slight decrease in credit for load resources (CLR). Therefore, the recommended IFRO for TI is **-459 MW/0.1 Hz**.

NERC requests that the recommended IFRO values calculated in this report in accordance with BAL-003-2 and shown in [Table ES.1](#) be approved for implementation in OY 2026. NERC, in collaboration with the RS, shall continue to monitor and evaluate the impacts on BPS reliability as a result of changes in IFRO values.

Table ES.1: Recommend IFROs for OY 2026					
	Eastern (EI)	Western (WI)	Texas (TI)	Québec (QI)	Units
MDF ⁴	0.420	0.280	0.405	0.947	Hz
RLPC ⁵	3,875	2,918	2,805	2,000	MW
CLR	N/A	N/A	947	N/A	MW
Calculated IFRO	-923	-1,042	-459	-211	MW/0.1 Hz
Recommended IFROs⁶	-923	-1,042	-459	-211	MW/0.1 Hz

⁴ The Procedure for ERO Support of Frequency Response and Frequency Bias Setting Standard, Version II, provided in the approved ballot for BAL-003-2, specifies that, "MDF is the Maximum Delta Frequency for the specific interconnection as determined in the 2017 Frequency Response Annual Analysis (FRAA)."

⁵ BAL-003-2, Attachment A specifies that Resource Loss Protection Criteria (RLPC) be based on the two largest potential resource losses in an interconnection. This value is required to be evaluated annually.

⁶ BAL-003-2 requires that the EI IFRO will be stepped down to its calculated value over three years. The maximum reduction is limited to 100 MW/0.10 Hz annually.

Introduction

This report contains the annual analysis, calculation, and recommendations for the IFRO for each of the four Interconnections of North America for the OY 2026 (December 2025 through November 2026). This analysis includes the following information:

- Statistical analysis of Interconnection frequency characteristics for the OYs 2020 through 2024 (December 1, 2019, through November 30, 2024)
- Analysis of frequency profiles for each Interconnection
- Calculation of adjustment factors from BAL-003-2 frequency response events

This year's frequency response analysis builds upon the work and experience from performing such analyses since 2013. As such, several important things should be noted about this report:

- The University of Tennessee–Knoxville FNET⁷ data used in the analysis has seen significant improvement in data quality, simplifying and improving annual analysis of frequency performance and ongoing tracking of frequency response events. In addition, NERC uses data quality checks to flag additional bad one-second data, including bandwidth filtering, least squares fit, and derivative checking.
- As with the previous year's analysis, all frequency event analysis uses sub-second data from the FNET system frequency data recorders (FDR). This eliminates the need for the CC_{ADJ} factor originally prescribed in the *2012 Frequency Response Initiative Report*² because the actual frequency nadir was accurately captured.
- The Frequency Response Analysis Tool⁸ is being used by the NERC Engineering and Security Integration group for frequency event tracking in support of the NERC Frequency Working Group and RS. The tool has streamlined interconnection frequency response analysis. The tool provides an effective means of determining frequency event performance parameters and generating a database of values necessary for calculation of adjustment factors.

This report contains numerous references to Value A, Value B, and Point C, which are defined in NERC BAL-003-2.¹ As such, it is important to understand the relationship between these variables and the basic tenets of primary and secondary frequency control.

The Arresting, Rebound, Stabilizing, and Recovery Periods of a frequency event following the loss of a large generation resource are shown in [Figure ES.1](#). Value A and Value B are average frequencies from t-16 to t-2 seconds and t+20 to t+52 seconds, respectively, as defined in NERC BAL-003-2. Point C is the lowest frequency experienced within the first 20 seconds following the start of a frequency event. A Point C' value may exist if frequency falls below the original Point C nadir or Value B after the end of the 20–52 second Stabilizing Period.

⁷ Operated by the Power Information Technology Laboratory at the University of Tennessee, FNET is a low-cost, quickly deployable GPS-synchronized wide-area frequency measurement network. High-dynamic accuracy FDRs are used to measure the frequency, phase angle, and voltage of the power system at ordinary 120 V outlets. The measurement data are continuously transmitted via the Internet to the FNET servers hosted at the University of Tennessee and Virginia Tech.

⁸ Developed by Pacific Northwest National Laboratory.

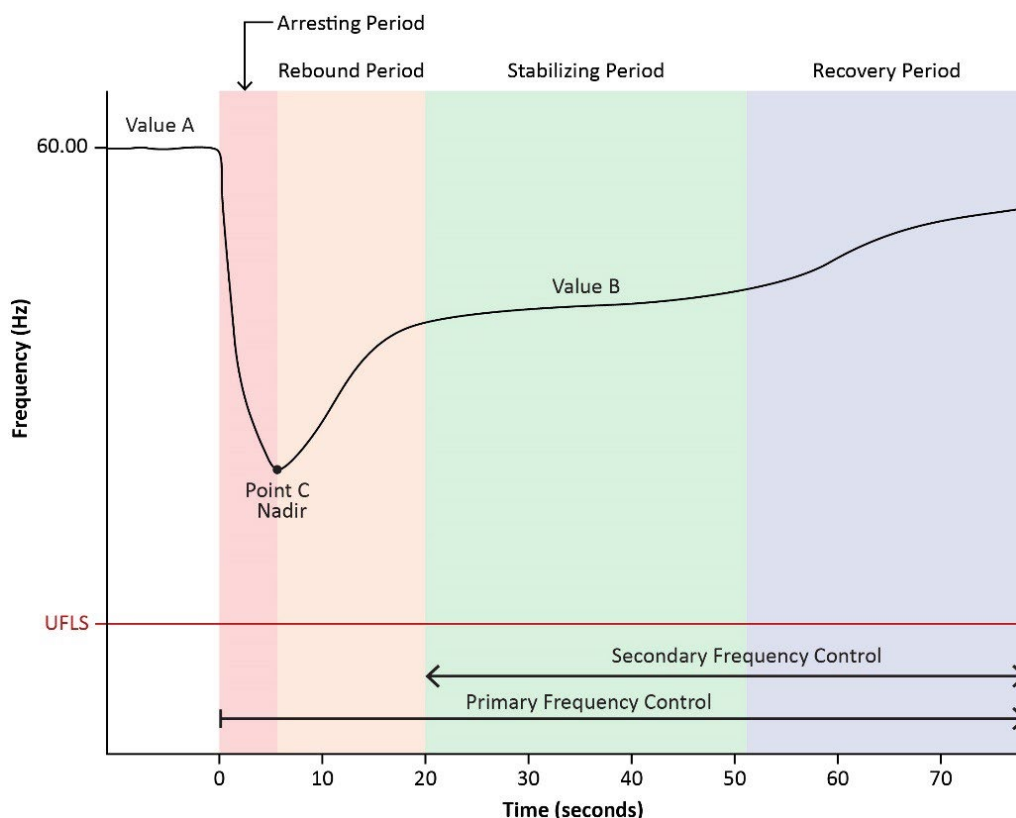


Figure ES.1: Primary and Secondary Frequency Control

Primary Frequency Control: This is the action by the Interconnection to arrest and stabilize frequency in response to frequency deviations and has three time components: the Arresting Period, Rebound Period, and Stabilizing Period. These terms are defined below:

- **Arresting Period:** This is the time from time zero (Value A) to the time of the nadir (Point C) and is the combination of system inertia, load damping, and the initial primary control response of resources acting together to limit the duration and magnitude of frequency change. It is essential that the decline in frequency is arrested during this period to prevent activation of automatic under-frequency load shedding (UFLS) schemes in the Interconnection.
- **Rebound Period:** This includes the effects of governor response in sensing the change in turbine speed as frequency increases or declines, causing an adjustment to the energy input of the turbine's prime mover. This can also be impacted by end-user customers or other loads that are capable of self-curtailment due to local frequency sensing and control during frequency deviations.
- **Stabilizing Period:** This is the third component of primary frequency control following a disturbance when the frequency stabilizes following a frequency excursion. Value B represents the interconnected system frequency at the point immediately after the frequency stabilizes primarily due to governor action but before the contingent control area takes corrective automatic generation control action.

Chapter 1: Interconnection Frequency Characteristic Analysis

Annually, NERC staff performs a statistical analysis (as detailed in the *2012 Frequency Response Initiative Report*)² of the frequency characteristics for each of the four Interconnections. That analysis is performed to monitor the changing frequency characteristics of the Interconnections and to statistically determine each Interconnection's starting frequency for the respective IFRO calculations. For this report's analysis, one-second frequency data^{9,10} from OYs 2020–2024 (December 1, 2019, through November 30, 2024) was used.

Frequency Variation Statistical Analysis

The 2025 frequency variation analysis was performed on one-second frequency data for 2020–2024 and is summarized in [Table 1.1](#). This variability accounts for items like time-error correction (TEC), variability of load, interchange, and frequency over the course of a normal day. It also accounts for all frequency excursion events.

The starting frequency is calculated and published in this report for comparison and informational purposes. Starting frequencies are evaluated annually and indicate no need to change the Maximum Delta Frequency (MDF) for OY 2026.

Table 1.1: Interconnection Frequency Variation Analysis 2020–2024				
Value	Eastern	Western	Texas	Québec
Number of Samples	157,022,983	157,001,174	155,530,339	152,296,105
Filtered Samples (% of total)	99.47	99.46	98.53	96.48
Expected Value (Hz)	59.999	60.000	59.999	59.999
Variance of Frequency (σ^2)	0.00027	0.00031	0.00029	0.00046
Standard Deviation (σ)	0.0165	0.0175	0.0169	0.0214
50% percentile (median) ¹³	59.999	59.999	60.004	59.998
Starting Frequency (F_{START}) (Hz)	59.971	59.970	59.970	59.964

The starting frequency is the fifth percentile of the five-year probability distribution of the respective interconnection frequency based on the statistical analysis, representing a 95% chance that frequencies will be at or above that value at the start of any frequency event. Since the starting frequencies encompass all variations in frequency, including changes to the target frequency during TECs, the need to expressly evaluate TEC as a variable in the IFRO calculation is eliminated.

[Figure 1.1](#), [Figure 1.2](#), [Figure 1.3](#), and [Figure 1.4](#) show the probability density function (PDF) of frequency for each Interconnection. The vertical black line indicates the fifth-percentile frequency; the interconnection frequency will statistically be greater than that value 95% of the time; this value is used as the starting frequency.

⁹ One-second frequency data for the frequency variation analysis is provided by UTK. The data is sourced from FDRs in each Interconnection. The median value among the higher-resolution FDRs is down-sampled to one sample per second, and filters are applied to ensure data quality.

¹⁰ Note regarding the EI median frequency that: with fast time error corrections the median value is around but slightly below 60 Hz. Without these corrections the median would be above 60 Hz.

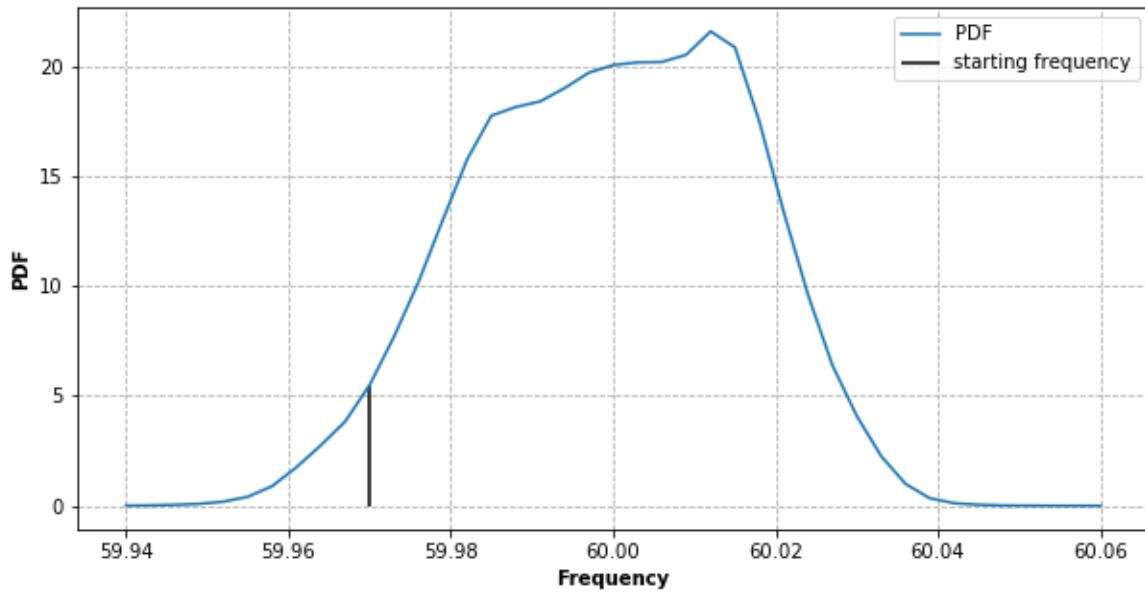


Figure 1.1: Eastern Interconnection 2020–2024 Probability Density Function of Frequency

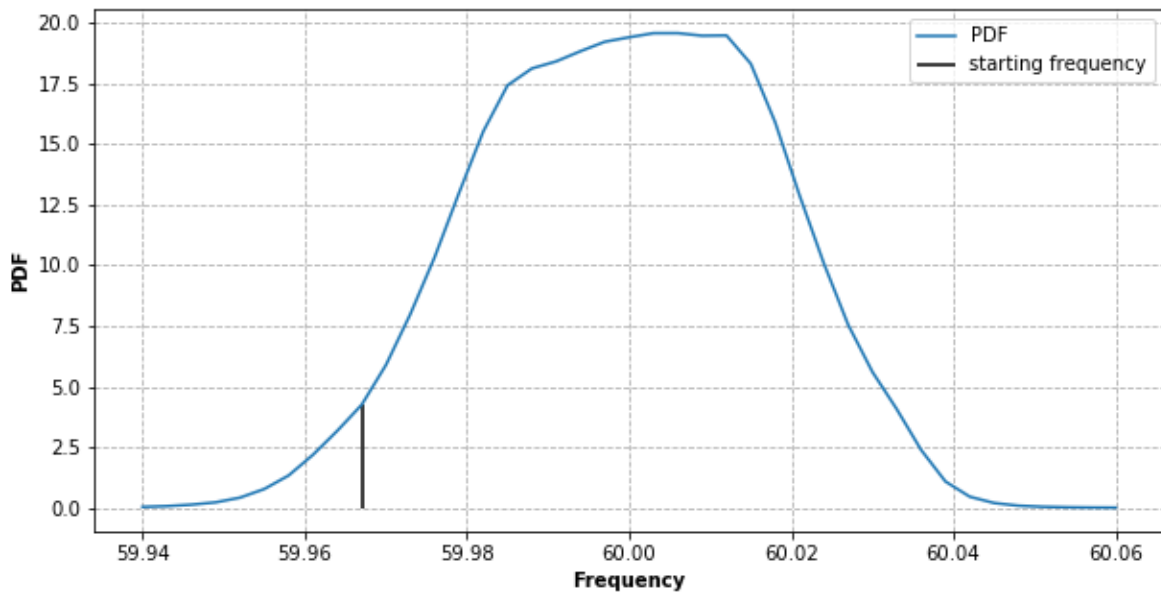


Figure 1.2: Western Interconnection 2020–2024 Probability Density Function of Frequency

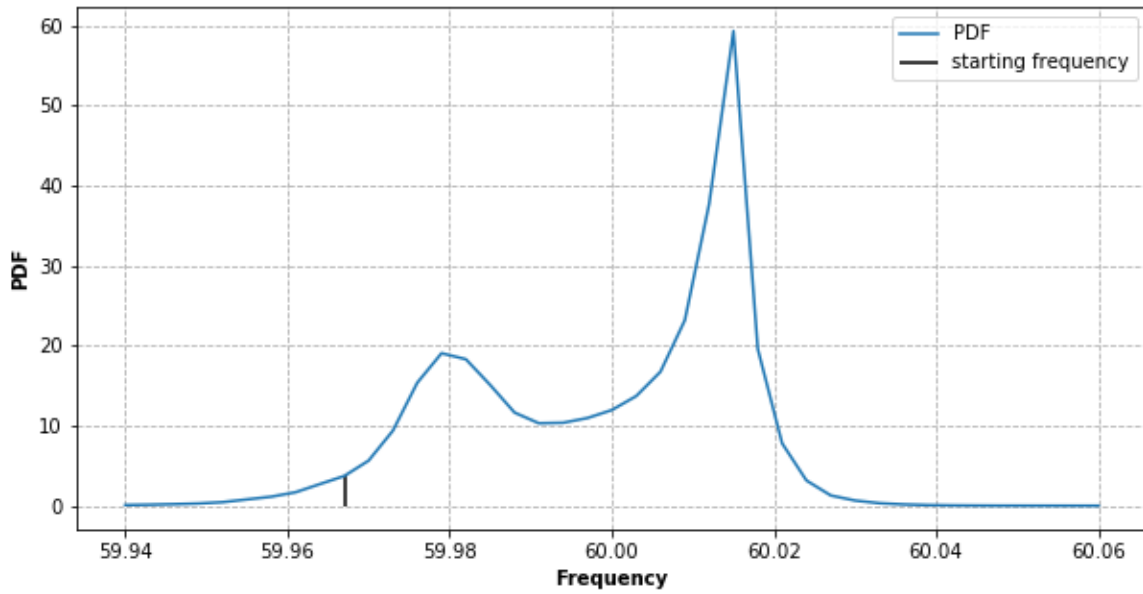


Figure 1.3: Texas Interconnection 2020–2024 Probability Density Function of Frequency

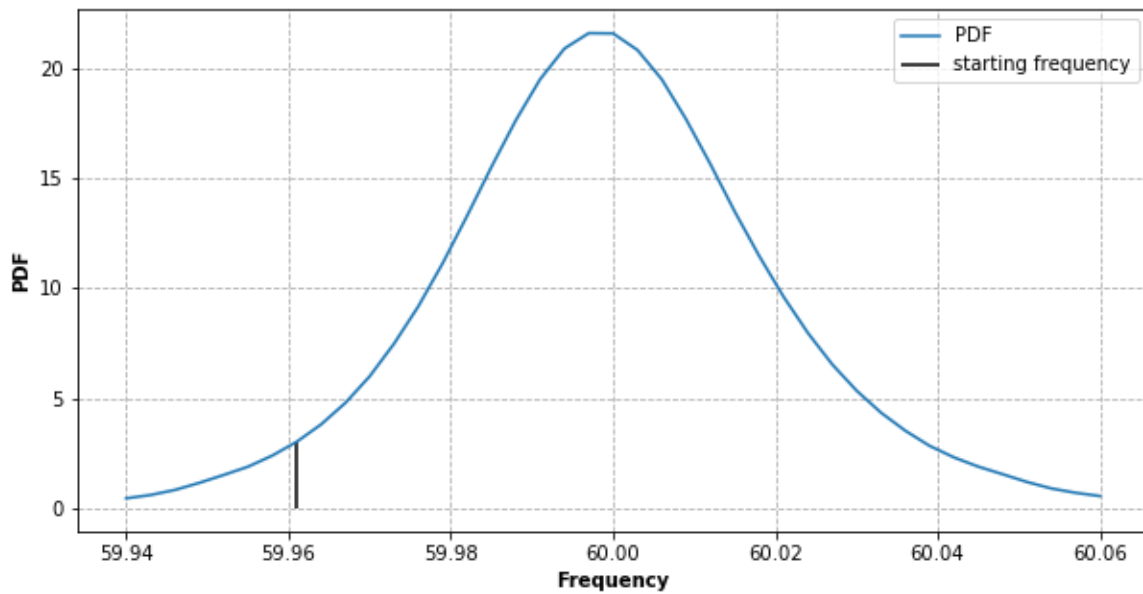


Figure 1.4: Québec Interconnection 2020–2024 Probability Density Function of Frequency

Figure 1.5 shows a comparison of the PDF for all Interconnections.

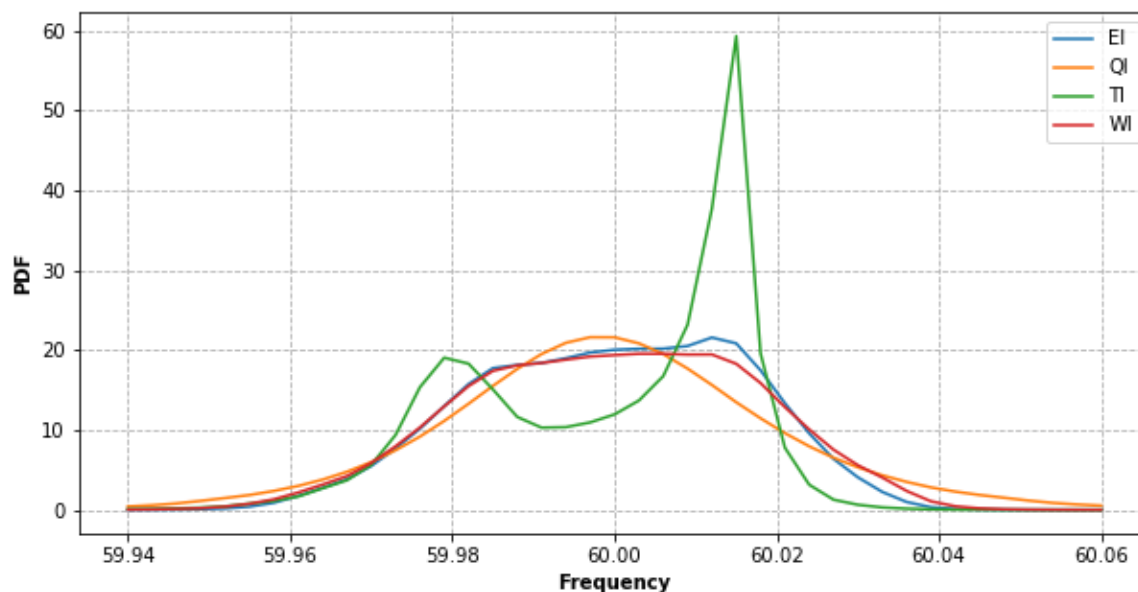


Figure 1.5: Comparison of 2020–2024 Interconnection Frequency PDFs

Variations in Probability Density Functions

The following is an analysis of the variations in probability density functions of the annual distributions of Interconnection frequency for years 2020–2024. Table 1.2 lists the standard deviation of the annual Interconnection frequencies.

Table 1.2: Interconnection Frequency Standard Deviation by Year					
Interconnection	2020	2021	2022	2023	2024
Eastern	0.0163	0.0164	0.0164	0.0167	0.0165
Western	0.0176	0.0174	0.0172	0.0177	0.0175
Texas	0.0174	0.0176	0.0169	0.0163	0.0169
Québec	0.0208	0.0223	0.0206	0.0220	0.0214

In the EI and WI, the standard deviation of Interconnection frequency appears to remain relatively stable for the past five years. In the TI, the standard deviation appears to have decreased slightly in recent years compared to 2020 and 2021. The standard deviation of Interconnection frequency in the QI appears to increase and decrease slightly year to year, decreasing slightly in 2024 compared to 2023. As standard deviation is a measure of dispersion of values around the mean value, the increasing standard deviations indicate reduced concentration around the mean value and less stable performance of the interconnection frequency. Comparisons of annual frequency profiles for each Interconnection are shown in Figure 1.6, Figure 1.7, Figure 1.8, and Figure 1.9.

Eastern Interconnection Frequency Characteristic Changes

The probability density function for the EI frequency characteristic in years 2020 to 2024 are shown in Figure 1.6. Statistical skewness (S)¹¹ increased in 2024 ($S = -0.20$) as compared to all four prior years ($S = -0.17, -0.16, -0.15$, and

¹¹ The skewness (S) is a measure of asymmetry of a distribution. A perfectly symmetric distribution has $S=0$. The sign indicates where a longer tail of the distribution is. The negatively skewed distribution has a longer left tail, and its curve leans to the opposite direction (to the right).

-0.17, respectively). NERC, in coordination with its technical committees, continues to evaluate this phenomenon and its impact, if any, on BPS reliability.

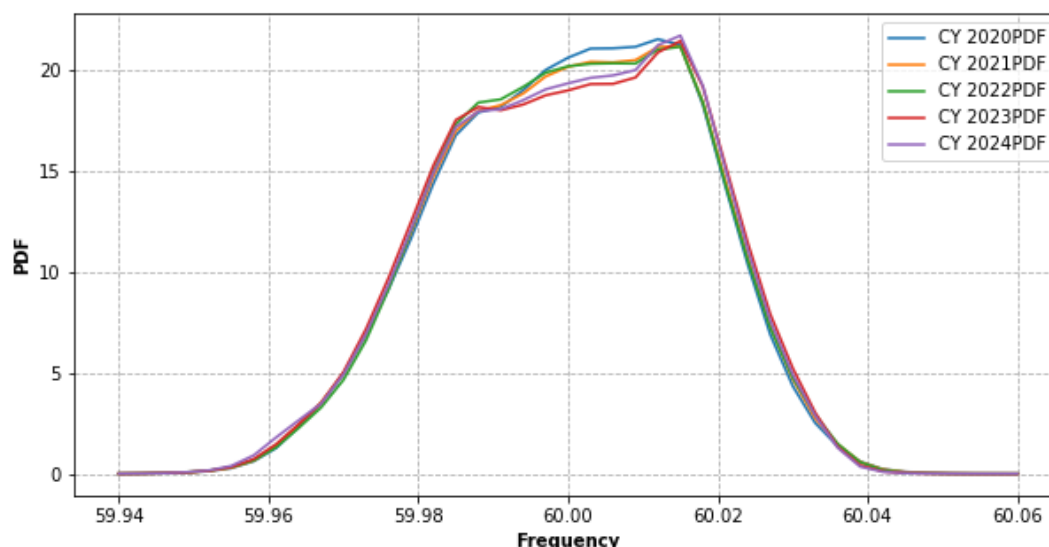


Figure 1.6: Eastern Interconnection Frequency Probability Density Function by Year

Western Interconnection Frequency Characteristic Changes

There was an observable change in the frequency distribution for the WI in 2022; the skewness decreased in magnitude from -0.11 in prior years to -0.09. However, the skewness was observed to increase in magnitude again in 2023 and 2024 to -0.13. Each of the past five years of probability density functions is shown below in [Figure 1.7](#)

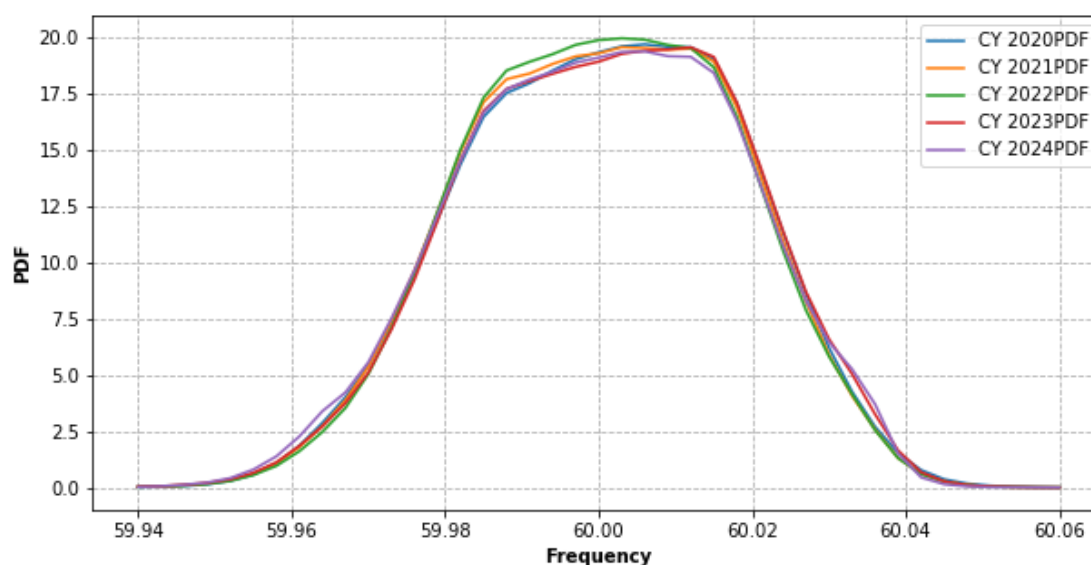


Figure 1.7: Western Interconnection Frequency Probability Density Function by Year

Algebraically, it means that the frequency values that are smaller than its mean are spread farther from the mean than the values greater than the mean or that there is more variability in lower values of the frequency than in higher values of the frequency.

Texas Interconnection Frequency Characteristic Changes

Standard BAL-001-TRE-1¹² requires all resources in the Texas Interconnection to provide proportional, non-step primary frequency response with a ± 17 mHz dead-band. As a result, if any time frequency exceeds 60.017 Hz, resources automatically curtail themselves. That has resulted in far less operation in frequencies above the dead-band since all resources, including wind and solar, are backing down. It is exhibited in [Figure 1.8](#) as a probability concentration around 60.015 Hz. Similar behavior is exhibited at the low dead-band of 59.983 Hz, but not as strong due to most wind and solar resources being operated at maximum output leading to an inability to increase output when frequency falls below the dead-band.

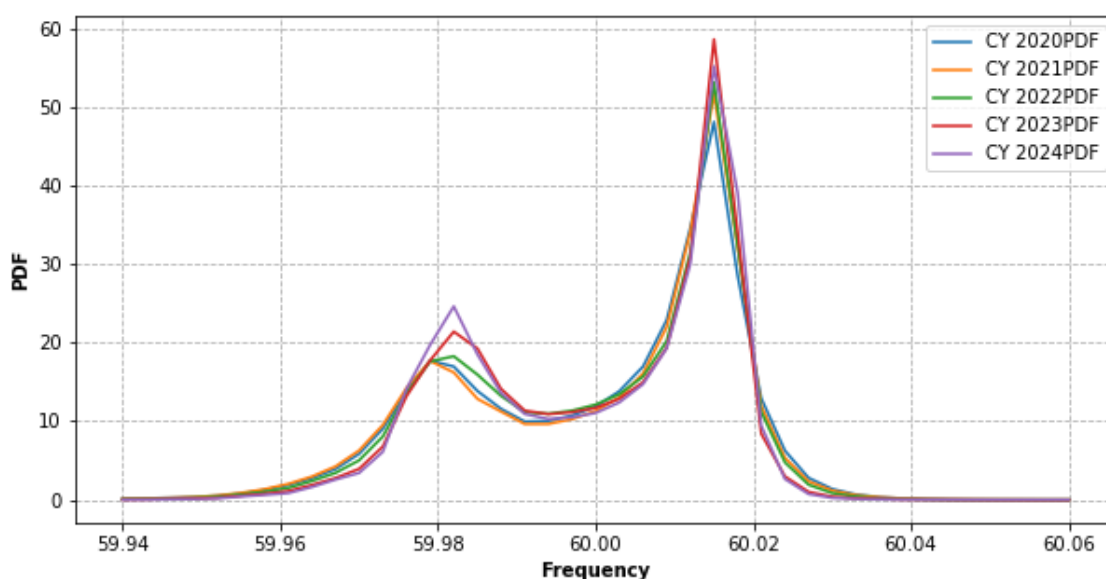


Figure 1.8: Texas Interconnection Frequency Probability Density Function by Year

Québec Interconnection Frequency Characteristic Changes

There were no major observable changes in the shape of the distribution for the QI as shown in [Figure 1.9](#).

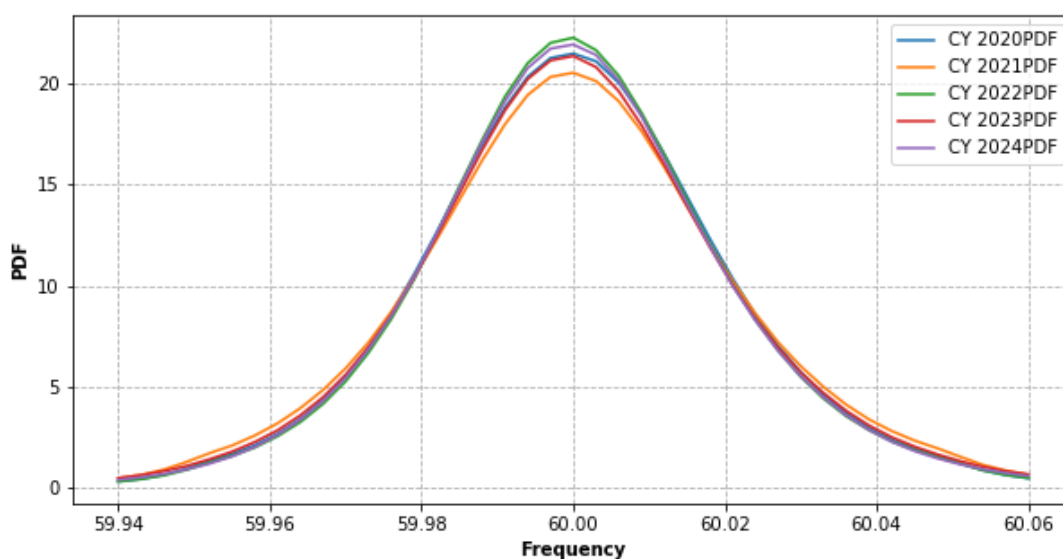


Figure 1.9: Québec Interconnection Frequency Probability Density Function by Year

¹² <http://www.nerc.com/pa/Stand/Reliability%20Standards/BAL-001-TRE-1.pdf>

Chapter 2: Determination of Interconnection Frequency Response Obligations

With this report, the IFRO calculations are determined by BAL-003-2. Previously, the calculation involved a multifaceted process that employed statistical analysis of past performance; analysis of the relationships between measurements of Value A, Point C, and Value B; and other adjustments to the allowable frequency deviations and resource losses used to determine the recommended IFROs. Refer to the *2012 Frequency Response Initiative Report* for additional details on the development of the IFRO and the adjustment calculation methods.² This report includes information that serves to transition from the old to the new method.

Tenets of IFRO

The IFRO is the minimum amount of frequency response that must be maintained by an Interconnection. Each Balancing Authority (BA) in the Interconnection is allocated a portion of the IFRO that represents its minimum annual median performance responsibility. To be sustainable, BAs susceptible to islanding may need to carry additional frequency-responsive reserves to coordinate with their UFLS plans for islanded operation.

A number of methods to assign the frequency response targets for each Interconnection can be considered. Initially, the following tenets should be applied:

- A frequency event should not activate the first stage of regionally approved UFLS systems within the Interconnection.
- Local activation of first-stage UFLS systems for severe frequency excursions, particularly those associated with delayed fault-clearing or in systems on the edge of an Interconnection, may be unavoidable.
- Other frequency-sensitive loads or electronically coupled resources may trip during such frequency events as is the case for photovoltaic (PV) inverters.
- It may be necessary in the future to consider other susceptible frequency sensitivities (e.g., electronically coupled load common-mode sensitivities). The Large Loads Task Force and other related RSTC subgroups are evaluating the risks to frequency stability from fault-induced load trips/transfers.

UFLS is intended to be a safety net to prevent system collapse due to severe contingencies. Conceptually, that safety net should not be utilized for frequency events that are expected to happen on a relatively regular basis. As such, the resource loss protection criteria were selected in accordance with BAL-003-2 to avoid violating regionally approved UFLS settings.

Interconnection Resource Loss Protection Criteria

BAL-003-2 introduced the Interconnection Resource Loss Protection Criteria (RLPC) to replace the Resource Contingency Protection Criteria used previously. It is based on resource loss in accordance with the following process:

Annually, BAs provide their potential two largest resource loss values and their largest resource loss due to an N-2 remedial action scheme (RAS) event. This will facilitate comparison between the existing Interconnection RLPC values and the RLPC values in use. This data submission will be needed to complete the calculation of the RLPC and IFRO.

BAs determine the two largest resource losses for the next OY based on a review of the following criteria and report only the two largest events from the three categories:

- The largest balancing contingency event(s) due to a single contingency identified using system models in terms of loss measured by megawatt loss in a normal system configuration (N-0) (An abnormal system configuration is not used to determine the RLPC).
- The largest facility(ies) in the BA area, regardless of shared ownership/responsibility
- The largest RAS resource loss(es) (if any) that are initiated by single (N-1) contingency events.

The BA provides these two numbers determined above as Resource Loss A and Resource Loss B in the FRS Form 1.

The BA should then provide the largest resource loss due to RAS operations (if any) that are initiated by a multiple contingency (N-2) event (RLPC cannot be lower than this value). If this RAS impacts more than a single BA, one BA is asked to take the lead and sum all resources lost due to the RAS event and provide that information.

The calculated RLPC should meet or exceed any credible N-2 resource loss event.

The host BA (or planned host BA) where jointly owned resources are physically located should be the only BA to report that resource. The full ratings of the resource, not the fractional shares, should be reported.

Direct current (DC) ties to asynchronous resources (such as DC ties between Interconnections, or the Manitoba Hydro Dorsey bi-pole ties to their northern asynchronous generation) should be considered as resource losses. DC lines, such as the Pacific DC Intertie, which ties two sections of the same synchronous Interconnection together, should not be reported. A single pole block with normal clearing in a monopole or bi-pole high-voltage direct current system is a single contingency.

Calculation of IFRO Values

The IFRO is calculated using the RLPC above ([Table 1 from BAL-003-2](#)).

$$IFRO = \frac{RLPC - CLR}{MDF * 10} \text{ MW/0.1Hz}$$

As specified in the *Procedure for ERO Support of Frequency Response and Frequency Bias Setting*¹³ standard, “MDF is the Maximum Delta Frequency for the specific interconnection as determined in the 2017 Frequency Response Annual Analysis (FRAA).” The BAL-003-2 revision alleviated the adverse impacts of an improving CB_R.

The IFRO for each Interconnection is calculated in this report in [Table 2.5](#); note that the calculated value for the EI IFRO, per BAL-003-2, was stepped down over three years with a reduction of IFRO not to exceed -100 MW/0.10 Hz per year in accordance with BAL-003-2. Collected RLPC data exceeded the estimate at the time BAL-003-2 balloted, and EI IFRO should meet the actual calculated value in only two OYs as a result. That determines the difference between the calculated EI IFRO in [Table 2.5](#) and the recommended IFRO shown in [Table ES.1](#) and [Table 2.7](#).

¹³ “Procedure for ERO Support of Frequency Response and Frequency Bias Setting Standard.” Available: https://www.nerc.com/pa/Stand/Project201701ModificationstoBAL00311/2019%20ERO%20Procedure%20document_October_2019_clean_from_last_posted_Project_2017_01_lka.pdf

Determination of Adjustment Factors

The C-to-B ratio (CB_R) is no longer used in the IFRO method and has been eliminated.

Adjustment for Primary Frequency Response Withdrawal (BC'_{ADJ})

Point C is normally the frequency nadir during the event; however, point C and the nadir may differ if the nadir occurs more than 20 seconds after the start of the event.¹⁴ This lower nadir is symptomatic of primary frequency response withdrawal or squelching by unit- or plant-level outer loop control systems. Withdrawal is most prevalent in the EI.

To track frequency response withdrawal in this report, the later-occurring nadir is termed Point C' (C prime), which is defined as occurring after the Value B averaging period and must be lower than either Point C or Value B.

Primary frequency response withdrawal is important depending on the type and characteristics of the generators in the resource dispatch, especially during light-load periods. Therefore, an additional adjustment to the maximum allowable delta frequency for calculating the IFROs was statistically developed. This adjustment is used whenever withdrawal is a prevalent feature of frequency events.

The statistical analysis is performed on the events with C' value lower than Value B to determine the adjustment factor BC'_{ADJ} to account for the statistically expected Point C' value of a frequency event. These results are correct for the influence of frequency response withdrawal on setting the IFRO. Table 2.1 shows a summary of the events for each Interconnection where the C' value was lower than Value B (averaged from T+20 through T+52 seconds) and those where C' was below Point C for OYs 2020 through 2024 (December 1, 2019, through November 30, 2024).

Interconnection	Number of BAL-003-2 Events Analyzed	C' Lower than B	C' Lower than C	Mean Difference Between B and C'	Standard Deviation	BC'_{ADJ} (95% Quantile)
EI	98	34	9	0.005	0.004	0.010
WI	88	48	0	N/A	N/A	N/A
TI	76	51	8*	N/A	N/A	N/A
QI	114	45	16*	0.028	0.062	0.035

* Load-loss events

The adjustment is not intended to be used for load-loss events. No generator loss events with C' lower than Point C were identified in the WI, TI, or QI. An adjustment factor is not warranted for interconnections that have no C' events from generator trips.

The adjustment factor of 10 mHz for the EI is necessary. Of the 98 frequency events analyzed in the EI, there were 34 events that exhibited a secondary nadir where Point C' was below Value B and 9 events where Point C' was lower than the initial frequency nadir (Point C). These secondary nadirs occur beyond 52 seconds after the start of the event,¹⁵ which is the time frame for calculating Value B.

Therefore, a BC'_{ADJ} is only needed for the EI; no BC'_{ADJ} is needed for the other three Interconnections. This will continue to be monitored moving forward to track these trends in C' performance.

¹⁴ The Procedure for ERO Support of Frequency Response and Frequency Bias Setting Standard defines Point C to occur within T+20 seconds.

¹⁵ The timing of the C' occurrence is consistent with outer-loop plant and unit controls, causing withdrawal of inverter-based resource frequency response.

Low-Frequency Limit

The low-frequency limits to be used for the IFRO calculations ([Table 2.2](#)) should be the highest step in the Interconnection for regionally approved UFLS systems. These values have remained unchanged since the *2012 Frequency Response Initiative Report*.

Table 2.2: Low-Frequency Limits (Hz)	
Interconnection	Highest UFLS Trip Frequency
EI	59.5
WI	59.5
TI	59.3
QI	58.5

The highest UFLS set point in the EI is 59.7 Hz in SERC-Florida Peninsula (FP), which was previously FRCC, while the highest set point in the rest of the Interconnection is 59.5 Hz. The SERC-FP 59.7 Hz first UFLS step is based on internal stability concerns and is meant to prevent the separation of the FP from the rest of the Interconnection. SERC-FP concluded that the IFRO starting point of 59.5 Hz for the EI is acceptable in that it imposes no greater risk of UFLS operation for an Interconnection resource-loss event than for an internal SERC-FP event.

Protection against tripping the highest step of UFLS does not ensure generation that has frequency-sensitive boiler or turbine control systems will not trip, especially in electrical proximity to faults or the loss of resources. Severe system conditions might drive the combination of frequency and voltage to levels that present some generator and turbine control systems to trip the generator. Similarly, severe rates-of-change occurring in voltage or frequency might actuate volts-per-hertz relays; this would also trip some generators, and some combustion turbines may not be able to sustain operation at frequencies below 59.5 Hz.

Distributed energy resources (DER), typically made up of inverter-based resources (IBR), may also be susceptible to extremes in frequency. New developments in DER standards reduce this concern, but some DERs that were energized before updated standards took effect may be more susceptible to extremes in frequency. This could become problematic in areas with a high penetration of IBRs.

Credit for Load Resources

The TI depends on contractual interruptible (an ancillary service) demand response that automatically trips at 59.7 Hz by under-frequency relays to help arrest frequency declines. A CLR is made for the resource contingency for the TI.

The amount of CLR available at any given time varies by different factors, including its usage in the immediate past. ERCOT performed statistical analysis on hourly available CLR over a one-year period from December 2023 through November 2024, like the approach used in prior FRAA reports.

Statistical analysis indicated that 947 MW of CLR is available 95% of the time. Therefore, a CLR adjustment of 947 MW is applied in the calculation of the TI IFRO as a reduction to the RLPC.

TI Credit for Load Resources

Prior to April 2012, the TI was procuring 2,300 MW of responsive reserve service, of which up to 50% could be provided by the load resources with under-frequency relays set at 59.70 Hz. Beginning April 2012, due to a change in market rules, the responsive reserve service requirement was increased from 2,300 MW to 2,800 MW for each hour, meaning load resources could potentially provide up to 1,400 MW of automatic primary frequency response.

Determination of Maximum Allowable Delta Frequencies

Because of the measurement limitation of the BA-level frequency response performance,¹⁶ IFROs must be calculated in “Value B space.” Protection from tripping UFLS for the Interconnections based on Point C, Value B, or any nadir occurring after Point C, within Value B, or after T+52 seconds must be reflected in the maximum allowable delta frequency for IFRO calculations expressed in terms comparable to Value B.

Table 2.3 shows the calculation of the maximum allowable delta frequencies for each of the Interconnections with one adjustment for the EI, BC'_{ADJ20} . However, the calculated MDF is not used in the IFRO calculation. The *Procedure for ERO Support of Frequency Response and Frequency Bias Setting Standard*⁴ states that the MDF is the Maximum Delta Frequency for the specific interconnection as determined in the 2017 FRAA, due to an improving CB_R . Dynamic simulation validation of the MDF and IFRO were last performed in the 2016 (for TI and WI) and 2017 (for EI) FRAA reports.

	EI	WI	TI	QI	Units
Starting Frequency	59.971	59.970	59.970	59.964	Hz
Minimum Frequency Limit	59.500	59.500	59.300	58.500	Hz
Base Delta Frequency	0.471	0.470	0.670	1.464	Hz
BC'_{ADJ20}	0.010	N/A	N/A	N/A	-
Calculated Max. Allowable Delta Frequency	0.461	0.470	0.670	1.464	Hz
Max. Delta Frequency Per <i>Procedure for ERO Support of Frequency Response and Frequency Bias Setting Standard</i>	0.420	0.280	0.405	0.947	Hz

¹⁶ Due to use of 1–6 second scan-rate in BA’s EMS systems to calculate the BA’s Frequency Response Measures for frequency events under BAL-003-2

Calculated IFROs

Table 2.4 shows the determination of IFROs for OY 2026 (December 2025 through November 2026) under standard BAL-003-2 based on a resource-loss equivalent to the recommended criteria in each Interconnection. The maximum allowable delta frequency values have already been modified to include the adjustments for the event nadir being below Value B (BC'_{ADJ}).

Table 2.4: Initial Calculation of OY 2026 IFROs					
	Eastern	Western	Texas	Québec	Units
Starting Frequency	59.971	59.970	59.970	59.964	Hz
Max. Delta Frequency Per <i>Procedure for ERO Support of Frequency Response and Frequency Bias Setting Standard</i>	0.420	0.280	0.405	0.947	Hz
Resource Loss Protection Criteria	3,875	2,918	2,805	2,000	MW
Credit for Load Resources	N/A	N/A	947	N/A	MW
Calculated IFRO using 2017 MDF	-923	-1042	-459	-211	MW/0.1 Hz
Recommended IFRO					
IFRO per <i>Procedure for ERO Support of Frequency Response and Frequency Bias Setting Standard</i>	-923	-1042	-459	-211	MW/0.10 Hz

Comparison to Previous IFRO Values

The IFROs were first calculated and presented in the *2012 Frequency Response Initiative Report*. **Table 2.5** compares the current IFROs and their key component values to those presented in the *2016 FRAA* report.

Table 2.5: Interconnection IFRO Comparison						
	OY 2016 Calc. ¹⁷	OY 2025 In Use ¹⁸	OY 2026 Calc. ¹⁹	OY 2016 Calc. to OY 2025 In Use Change	OY 2025 In Use to 2026 Calc. Change	Units
Eastern Interconnection						
Starting Frequency	59.974	59.971	59.971	-0.003	0.000	Hz
Max. Allowable Delta Frequency	0.443	0.420	0.420	-0.023	0.000	Hz
Resource Contingency Protection Criteria	4500	3,875	3,875	-625	0	MW
Credit for Load Resources	0	0	0	0	0	MW
Absolute Value of IFRO	1015	923	923	-92	0	MW/0.1 Hz
Western Interconnection						
Starting Frequency	59.967	59.970	59.970	0.003	0.000	Hz
Max. Allowable Delta Frequency	0.292	0.280	0.280	-0.012	0.000	Hz
Resource Loss Protection Criteria	2626	2918	2918	292	0	MW
Credit for Load Resources	0	0	0	0	0	MW

¹⁷ Calculated in the 2015 FRAA report. Average frequency values were for OYs 2012–2014.

¹⁸ Calculated in the 2024 FRAA report. Average frequency values were for OYs 2019–2023.

¹⁹ Calculated in the 2025 FRAA report. Average frequency values were for OYs 2020–2024.

Table 2.5: Interconnection IFRO Comparison

	OY 2016 Calc. ¹⁷	OY 2025 In Use ¹⁸	OY 2026 Calc. ¹⁹	OY 2016 Calc. to OY 2025 In Use Change	OY 2025 In Use to 2026 Calc. Change	Units
Absolute Value of IFRO	858	1042	1042	184	0	MW/0.1 Hz
Texas Interconnection						
Starting Frequency	59.966	59.970	59.970	0.004	0.000	Hz
Max. Allowable Delta Frequency	0.411	0.405	0.405	-0.006	0.000	Hz
Resource Loss Protection Criteria	2750	2805	2805	55	0	MW
Credit for Load Resources	1,181	962	947	219	15	MW
Absolute Value of IFRO	381	455	459	74	4	MW/0.1 Hz
Quebec Interconnection						
Starting Frequency	59.969	59.964	59.964	-0.005	-0.001	Hz
Max. Allowable Delta Frequency	0.948	0.947	0.947	-0.001	0.000	Hz
Resource Loss Protection Criteria	1700	2000	2000	300	0	MW
Credit for Load Resources	0	0	0	0	0	MW
Absolute Value of IFRO	179	211	211	32	0	MW/0.1 Hz

Key Findings

Table 2.6 shows a comparison of mean Value A, mean Value B, and mean Point C for qualifying BAL-003-2 events that are illustrative of Interconnection performance over the previous OY and as compared to the 2016 OY in which the IFRO values were frozen. Loss-of-load events have been excluded from the data in **Table 2.6**. The Eastern, Western, and Texas Interconnections show an increase in mean Value B and a decrease in the mean (A-B), indicating improved performance during the stabilizing period of frequency events. Québec showed an increase in mean Value B up until OY 2024, where it declined slightly. Western, Texas, and Québec Interconnections show an increase in mean Point C as well as a decrease or no change in mean (A-C), indicating improved performance during the arresting period of frequency events. The Eastern Interconnection shows no change in mean Value C as well as a slight decrease in mean Value A-C.

**Table 2.6: Year over Year Comparison Value A, Value B, and Point C
(Loss of Load Events Excluded)**

	OY2016	OY2024	OY2025	Difference OY 2024–2016	Difference OY 2025–2024
Eastern Interconnection					
Mean Value A (Hz)	59.998	60.001	59.998	0.003	-0.003
Mean Value B (Hz)	59.947	59.957	59.958	0.010	0.001
Mean Point C (Hz)	59.947	59.948	59.948	0.001	0.000
Mean A – B (Hz)	0.051	0.044	0.040	-0.007	-0.004
Mean A – C (Hz)	0.051	0.053	0.050	0.002	-0.003
Western Interconnection					
Mean Value A (Hz)	60.000	59.998	59.999	-0.002	0.001
Mean Value B (Hz)	59.923	59.952	59.960	0.029	0.008
Mean Point C (Hz)	59.887	59.901	59.915	0.014	0.014
Mean A – B (Hz)	0.076	0.046	0.039	-0.030	-0.007
Mean A – C (Hz)	0.112	0.097	0.083	-0.015	-0.014

**Table 2.6: Year over Year Comparison Value A, Value B, and Point C
(Loss of Load Events Excluded)**

	OY2016	OY2024	OY2025	Difference OY 2024–2016	Difference OY 2025–2024
Texas Interconnection					
Mean Value A (Hz)	59.996	60.000	60.010	0.004	0.010
Mean Value B (Hz)	59.889	59.931	59.955	0.042	0.024
Mean Point C (Hz)	59.840	59.866	59.912	0.026	0.046
Mean A – B (Hz)	0.107	0.070	0.055	-0.037	-0.015
Mean A – C (Hz)	0.156	0.134	0.098	-0.022	-0.036
Quebec Interconnection					
Mean Value A (Hz)	60.003	59.998	60.003	-0.005	0.005
Mean Value B (Hz)	59.843	59.952	59.931	0.109	-0.021
Mean Point C (Hz)	59.433	59.901	59.652	0.468	-0.249
Mean A – B (Hz)	0.160	0.046	0.072	-0.114	0.026
Mean A – C (Hz)	0.570	0.097	0.351	-0.473	0.254

Recommended IFROs for OY 2026

Consistent with the requirements of BAL-003-2, the IFRO values shown in [Table 2.7](#) for OY 2026 (December 2025 through November 2026) are recommended as follows:

Table 2.7: Recommended IFROs for OY 2026

	EI	WI	TI	HQ	Units
MDF ⁴	0.420	0.280	0.405	0.947	Hz
RLPC ²⁰	3875	2918	2805	2000	MW
CLR	0	0	947	0	MW
Calculated IFRO	-923	-1042	-459	-211	MW/0.1 Hz
Recommended IFRO ²¹	-923	-1042	-459	-211	MW/0.1 Hz

²⁰ BAL-003-2, Attachment A specifies that Resource Loss Protection Criteria (RLPC) be based on the two largest potential resource losses in an interconnection. This value is required to be evaluated annually.

²¹ BAL-003-2 requires that the EI IFRO will be stepped down to its calculated value over three years. The maximum reduction is limited to 100 MW/0.10 Hz annually.

Chapter 3: Dynamics Analysis of Recommended IFROs

Because the IFROs for the EI, WI, and TI have only been calculated upon issue of this report, they have not been changed as governed by BAL-003-2. Additional dynamic validation analyses were not done for this report.

Refer to the dynamics validation in the 2017 FRAA²² report for details. No analysis was performed for the QI.

Further supporting dynamic studies accompanied the development and filing of BAL-003-2.

²² https://www.nerc.com/comm/OC/Documents/2017_FRAA_Final_20171113.pdf